

Problem

Submissions

Leaderboard

Discussions

ALGORITHM

You are given a directed flow network consisting of V vertices and E directed edges. Each edge has a capacity that represents the maximum amount of flow that can pass through it.

The source vertex is 0 and the sink vertex is $V-1$.

Your task is to find the maximum possible flow from the source to the sink using the Ford—Fulkerson algorithm.

Input Format

The first line contains two space-separated integers: V — number of vertices E — number of directed edges Each of the next E lines contains three space-separated integers: u — starting vertex v — ending vertex capacity — capacity of the edge

Constraints

$2 \leq V \leq 100$ $1 \leq E \leq 1000$ $0 \leq u, v < V$ $u \neq v$ $1 \leq \text{capacity} \leq 10^6$

Output Format

Print a single integer representing the maximum flow from vertex 0 to vertex $V-1$.

Sample Input 0

```
6 9
0 1 16
0 2 13
1 2 10
2 1 4
1 3 12
3 2 9
2 4 14
4 3 7
3 5 20
```

Sample Output 0

```
19
```

Change Theme

Language

Java 15

```
1 import java.util.*;
2
3 class Solution {
4
5     static int V;
6
7     static boolean bfs(int[][] graph, int[] parent,
8         boolean[] visited = new boolean[V];
9         Queue<Integer> q = new LinkedList<>();
10
11         q.add(source);
12         visited[source] = true;
13         parent[source] = -1;
14
15         while (!q.isEmpty()) {
16             int u = q.poll();
17
18             for (int v = 0; v < V; v++) {
19                 if (!visited[v] && graph[u][v] > 0) {
20
21                     q.add(v);
22                     parent[v] = u;
23                     visited[v] = true;
24
25                     if (v == sink) {
```

Line: 89 Col: 2

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Test case 0

Compiler Message

Success